The Maintenance Society: Coordination of Preventive Work in Small Language Model Agents

Abstract

Groups that share equipment must divide their time between producing output and maintaining the equipment. Production offers an immediate reward, while maintenance preserves future opportunities to produce. We introduce The Maintenance Society, a workshop simulation in which six small language model agents choose between production and maintenance over 30 rounds. Production earns individual points and increases machine wear. Maintenance earns no points and reduces wear. Breakdowns interrupt production for two rounds. We compare two small open-weight models across four conditions that vary the display of maintenance totals and the availability of discussion. Of 80 scheduled episodes, 69 finished the workshop. Displaying maintenance totals increased Phi's mean output from 336.0 to 394.5 points without discussion, alongside a reduction in maintenance from 68.0 to 48.5 actions. The same dashboard change did not consistently improve Qwen's output. Discussion did not consistently improve coordination. Completed Phi discussion workshops shared maintenance more evenly but wasted over 84% of their supplied maintenance capacity. Most of their final maintenance actions followed an initial intention to produce. Recognition results varied across conditions and were limited by incomplete responses. These findings show that contributing to upkeep, sharing its burden, and performing it when needed are distinct aspects of coordination.

Keywords large language models · multi-agent systems · cooperation · preventive maintenance

1 Introduction

LLM-based agents can operate in settings with limited resources, such as edge devices, where compute, memory, and power restrict the models that can be deployed. When agents share equipment or infrastructure, they must also coordinate the work needed to keep those resources available. A group may agree that maintenance matters while still performing too little, assigning it to the same participant, or maintaining equipment that does not need attention.

The difficulty comes from the timing of costs and benefits. An agent that produces an item receives an immediate reward. An agent that maintains the shared machine gives up that reward, although its action can preserve production opportunities for everyone. A useful division of work must therefore address both the amount of maintenance and the distribution of its cost. Equal participation alone does not establish that the group is using its time well.

GovSim studies whether LLM agents can sustain a shared resource by limiting how much they extract [1]. Agents preserve future access through restraint, and resource collapse prevents further extraction. The Energy Society studies a different pressure: generating tokens consumes energy that agents need to remain active [2]. It compares objectives of maximising individual or group energy and examines how those objectives change donations and job allocation.

The Maintenance Society focuses on allocating a separate upkeep task. Agents choose between earning production points and maintaining a machine that everyone uses. Maintenance requires giving up a production opportunity, so the group must decide who bears that cost and when the work is needed. Machine breakdowns cause temporary shutdowns followed by automatic repair. This allows us to observe maintenance choices across repeated periods of operation and recovery.

Our experimental contribution is to test whether displaying each agent's maintenance total changes that allocation while the individual objective and rewards remain fixed. We compare the two dashboards with and without discussion. Agents receive the full action history in all conditions, so the intervention supplies a maintenance summary rather than new information. We measure production, maintenance concentration, and unused maintenance capacity together. This comparison distinguishes groups that share useful upkeep from groups that share excessive maintenance equally, a question not directly tested by the extraction and objective comparisons above.

We ask two questions. Does displaying maintenance totals change production and the distribution of maintenance? Does discussion help agents coordinate the amount and timing of that work? We also examine whether final actions

follow stated maintenance intentions and whether agents who maintain more receive more recognition after the workshop.

The study contributes a small environment with explicit action costs and exact wear accounting, together with a four-condition comparison using two locally runnable models. Deterministic reference policies establish that high production and equal maintenance are achievable under the same rules. The observed LLM groups often fall short because they maintain unnecessarily or fail to maintain at the required time. We present these results as evidence about the tested configurations rather than a general ranking of language models.

2 The Maintenance Society

2.1 Actions and machine wear

The Maintenance Society contains six agents and one shared machine. An episode consists of 30 rounds. During an operational round, each agent chooses one action.

Produce. The agent produces one item, earns three personal points, and adds one unit of machine wear.

Maintain. The agent earns no production points and supplies six units of wear reduction. Wear cannot fall below zero, so some of this capacity may go unused.

The simulator collects all six final actions before applying them together. Let p_t be the number of producers and m_t the number of maintainers in round t. Since each agent chooses one action, $p_t + m_t = 6$. Starting from $W_0 = 12$, the wear update in an operational round is:

$$\begin{aligned} W_t &= \max(0, W_{t-1} + p_t - 6m_t) \\ &= \max(0, W_{t-1} + 6 - 7m_t). \end{aligned} \tag{1}$$

A breakdown occurs if the updated wear is at least 30. Producers keep the points earned during that round. The next two rounds are closed: agents take no actions and earn no points. At the end of the second closed round, automatic repair resets wear to 12. Closed rounds count toward the 30-round episode. A breakdown in round 30 is recorded even though its shutdown would occur after the measured episode.

For example, consider a round starting at wear 24. If all six agents produce, the group earns 18 points and wear reaches 30. The machine then closes for two rounds. If one agent maintains while five produce, the group earns 15 points and wear falls to 23. That maintainer gives up three immediate points while keeping the machine available for the next round.

2.2 Agent observations and discussion

Agents receive their identity, the current round, the 30-round limit, the number of rounds remaining, and the current machine wear. They also receive each participant's cumulative production points and the complete history of completed rounds. Each history entry contains starting and ending wear, the six actions, and any breakdown or automatic repair. Depending on the condition, the dashboard also lists each participant's cumulative maintenance count.

In discussion conditions, each agent first submits a message and an intended action. The six proposals are collected independently and revealed together. Each agent then chooses its own final action after reading the proposals. It cannot observe the other agents' final choices before committing. Intentions are nonbinding, so an agent may change its action after reading the messages. Fig. 1 shows the interaction sequence.

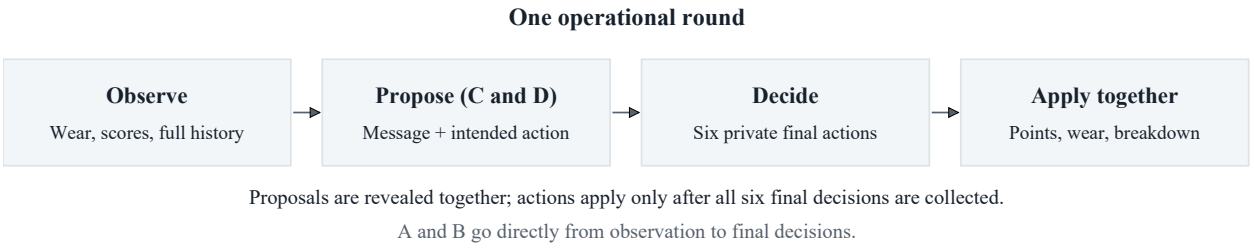


Figure 1: Interaction sequence for one operational round. Conditions A and B skip the proposal stage. Final actions apply together after all six agents have submitted valid decisions.

Suppose two agents both propose maintenance. After reading the proposals, one might switch to production while the other maintains. This would remove duplicate effort. Alternatively, several agents might switch from production to maintenance and create redundant work. We record both the intended and final actions to distinguish these patterns.

The discussion protocol permits messages of up to 60 words. A proposal input contains up to five previous discussion rounds. A final-decision input contains the current proposals and up to four previous discussion rounds. The full completed action history remains available at both stages. We use separate observations for each agent; identities do not share private model conversations.

2.3 Recognition after the workshop

After round 30, each agent independently distributes 100 integer recognition points among the other five agents. Self-recognition is excluded. Agents receive the completed workshop history and assess their peers' contributions. Recognition does not change production points or influence subsequent workshop actions, because the episode has already ended. We use it as an additional measure of which participants receive credit for the completed work.

3 Experimental setup

3.1 Models and conditions

We evaluate Qwen3-4B-Instruct-2507 and Phi-4-mini-instruct, using Q4_K_M quantisation for both. We refer to them as Qwen and Phi. Each episode contains six identities supplied by the same model configuration. We compare homogeneous groups rather than mixing model families within an episode.

Both models run locally through Ollama on a machine with 16 GB RAM and an RTX 3060 with 6 GB VRAM. We load one model and issue one request at a time. Requests run sequentially for hardware reasons, while the simulator applies the six chosen actions simultaneously. Generation uses an 8,192-token context, temperature 0.7, top-p 0.9, top-k 40, and repetition penalty 1.0. Output limits are 64 tokens for final decisions and 256 tokens for proposals and recognition. Qwen's thinking option is disabled.

The four conditions form a two-by-two comparison of dashboard content and discussion (Table 1). Production rewards, machine rules, and the objective of maximising individual production points remain the same.

Table 1: Experimental conditions. All conditions provide production totals and the full action history. B and D additionally summarise maintenance counts.

Condition	Dashboard totals	Discussion
A	Production points	No
B	Production points and maintenance counts	No
C	Production points	Yes
D	Production points and maintenance counts	Yes

An agent in A or C can count maintenance from the history. In B and D, the dashboard supplies that count. We do not hide maintenance records in A or C, and we do not pay agents for maintenance in B or D.

We schedule ten episodes per model and condition, giving 80 attempts. Episode seeds and agent display orders are matched across conditions and models. The order of conditions is shuffled within each episode index. We treat the episode as the unit of analysis because the six participants act within a shared environment.

3.2 Reference policies

We compare observed performance with five reference policies. Each policy uses a fixed rule to select the agents' actions, without calling an LLM. They show what happens under different approaches to maintenance. The rules below apply during operational rounds; agents take no actions during a shutdown.

Always produce. All six agents produce in each operational round. Nobody maintains. This policy shows the output and breakdowns that result when the group neglects maintenance.

One rotating maintainer per round. One agent maintains while the other five produce, regardless of wear. Agent 1 maintains in round 1, agent 2 in round 2, and so on. The rotation repeats after agent 6. This policy shares maintenance equally through a regular schedule.

Threshold 24: designated maintainer. If a round starts at wear 24 or above, agent 1 maintains and the other five produce. Below wear 24, all six produce. The same agent bears the entire maintenance burden.

Threshold 24: rotating maintainer. This policy uses the same wear threshold, but rotates the maintenance assignment. Agent 1 handles the first maintenance action, agent 2 the next, and so on. The assignment advances only when maintenance occurs. This policy combines maintenance when needed with shared responsibility.

Terminal aware threshold. Through round 29, this policy follows the designated maintainer threshold rule. In round 30, all six agents produce regardless of wear. It uses the known episode ending to gain production points even if the final action causes a breakdown. Any resulting shutdown falls outside the measured episode.

Table 2 reports the outcomes of these policies under the same workshop rules used for the LLM agents.

Table 2: Reference outcomes over 30 rounds. Maintenance counts individual agent actions. Lost rounds are closed rounds within the episode. The terminal-aware policy permits a breakdown in round 30.

Policy	Points	Maint. actions	Break-downs	Lost rounds
Always produce	324	0	6	12
One rotating maintainer	450	30	0	0
Designated threshold	468	24	0	0
Rotating threshold	468	24	0	0
Terminal-aware threshold	471	23	1	0

A designated maintainer performs all 24 maintenance actions and earns 18 points, while the other agents earn 90 each. Rotating responsibility at the same threshold gives each agent four maintenance actions and 78 points. Both groups earn 468 points without a breakdown. Thus the task permits equal maintenance and high production at the same time.

The terminal aware policy earns 471 points by making all six agents produce in round 30. Wear rises from 25 to 31, causing a breakdown. The resulting shutdown would close rounds 31 and 32, which fall outside our 30-round experiment. The policy therefore gains three points over the regular threshold policies without losing any production rounds within the measured episode.

3.3 Episodes included in the analysis

We analyse production, maintenance, and breakdowns using episodes that finish all 30 workshop rounds. We include a finished workshop even if recognition is incomplete. Recognition analysis requires valid allocations from all six agents. We exclude unfinished workshop totals from comparisons of 30-round outcomes and report the available sample size beside each result.

Outputs must follow the required response format. An invalid response receives one correction request. A second invalid response ends that attempt; the program does not substitute an action. Of the 80 attempts, 69 finish the workshop and 50 also finish recognition. Table 3 lists the samples used below. The smaller Phi discussion samples limit comparisons involving those conditions.

Table 3: Finished workshops and complete recognition sets, out of ten scheduled attempts per row. Workshop analyses use the first count; recognition analyses use the second.

Model	Condition	Finished workshops	Complete recognition
Qwen	A	10	9
Qwen	B	10	10
Qwen	C	10	10
Qwen	D	10	9
Phi	A	10	3
Phi	B	10	5
Phi	C	4	2
Phi	D	5	2

4 Evaluation Metrics

We measure production, the distribution and usefulness of maintenance, and the relation between discussion and action. These outcomes address different parts of the coordination problem. We report episode means and sample standard deviations.

To compare two conditions, we pair episodes with the same index and include a pair only if both workshops finish all 30 rounds. We calculate the difference within each pair, then average those differences. To estimate uncertainty in that average, we randomly resample the observed differences 20,000 times, allowing repeated selections, and calculate an average for each resample. We report the middle 95% of these averages as a descriptive bootstrap interval. Supplementary Table S1 provides the paired production differences and their intervals.

This procedure reuses the recorded observations; it does not add new experimental episodes or recover the missing final scores. Some Phi comparisons contain only three to five finished pairs. Their estimates remain tentative because we do not know how the unfinished episodes would have ended.

4.1 Production and breakdowns

Group points are three times the total number of production actions. Let M be the number of maintenance actions and L the number of closed rounds in a finished episode. There are 180 possible agent actions across six agents and 30 rounds, giving:

$$\text{Group points} = 540 - 3M - 18L. \tag{2}$$

Each maintenance action replaces one three point production opportunity. Each closed round removes six such opportunities. The value 540 is an arithmetic starting point; it is not achievable under the wear rules. **Score gap** is the highest individual production score minus the lowest.

Breakdowns count operational rounds ending at wear 30 or above. **Lost rounds** count the resulting closed rounds within the episode. They differ near the episode boundary: a breakdown in round 29 closes only round 30 within the measurement window, while a breakdown in round 30 causes no measured lost round.

4.2 Maintenance distribution and waste

Maintenance actions count how often agents choose maintenance. Three agents maintaining in one round count as three actions. **Maintenance concentration** is the largest individual share of the group's maintenance:

$$\text{Concentration} = \frac{\max_i M_i}{M}. \tag{3}$$

Here M_i is agent i's maintenance count. A value of $1/6$, or about 0.167, represents equal sharing; 1 means one agent performed all maintenance. We leave concentration undefined when $M = 0$.

Wasted Wear Reduction (WWR) measures maintenance capacity that cannot remove wear because wear cannot fall below zero:

$$\mathrm{WWR} = \sum_t \max(0, 6m_t - W_{t-1} - p_t). \tag{4}$$

The sum covers operational rounds. For example, starting at wear two, two maintainers supply 12 units of reduction while four producers add four units of wear. Only six units need removal, so six units of capacity are wasted.

We also report **unused maintenance capacity**, the percentage of potential wear reduction that went unused. For each model and condition, we combine its n finished workshops:

$$\begin{aligned} &\text{Unused capacity } (\%) \\ &\quad = 100 \times \frac{\sum_{e=1}^{n} \mathrm{WWR}_e}{6 \sum_{e=1}^{n} M_e}. \end{aligned} \tag{5}$$

Here e identifies an episode, WWR_e is its wasted wear reduction, and M_e is its number of maintenance actions. Each action supplies six units of capacity. The numerator sums wasted capacity across the finished workshops; the denominator sums their supplied capacity.

4.3 Stated intentions and recognition

Maintenance-intention fulfilment is the fraction of stated maintenance intentions followed by a final maintenance action. For each model and condition, we add the fulfilled maintenance intentions across finished episodes and divide by the total declared maintenance intentions. An episode with no maintenance intentions contributes no declarations. We also count final maintenance actions that follow an initial production intention. These measures locate changes between proposal and decision; they do not establish why an agent changed its action.

Received recognition is the sum of points an agent receives from its five peers. In each episode with complete recognition, we calculate the Spearman correlation between the six maintenance counts and the six received totals. This compares the two rankings. A positive value means agents who maintain more tend to receive more recognition; a negative value means they tend to receive less. We assign average ranks to ties. If all maintenance counts or all received totals are equal, the correlation is undefined. We report the mean of the defined episode correlations and the number available. Maintenance count measures effort spent, so this association does not establish whether recognition tracks useful maintenance.

5 Results

The main coordination problem is to keep the machine working while leaving enough agents free to produce. Across the finished workshops, groups often perform more maintenance than they need, and some still suffer breakdowns. The dashboard and discussion change how agents allocate this work, but neither produces an efficient schedule across both model configurations.

5.1 Production and breakdowns

The clearest difference between the models concerns how they lose production opportunities. Phi groups without discussion keep the workshop open throughout the episode, but spend many actions maintaining it. Qwen groups incur both the cost of maintenance and the loss of production during shutdowns. Table 4 summarises these outcomes, and Fig. 2 compares output with the failure free reference.

Table 4: Performance across finished 30-round workshops. Points include the sample standard deviation; the other outcomes are means. *M* denotes maintenance actions and *L* denotes closed rounds within the episode. Phi C and D include four and five finished workshops, respectively.

Model	Condition	*n*	Points (mean ± SD)	*M*	Breakdowns	*L*
Qwen	A	10	362.7 ± 62.5	44.7	1.5	2.4
Qwen	B	10	341.4 ± 45.5	39.8	2.5	4.4
Qwen	C	10	328.8 ± 35.5	41.0	2.5	4.9
Qwen	D	10	321.6 ± 61.4	46.4	2.4	4.4
Phi	A	10	336.0 ± 31.5	68.0	0.0	0.0
Phi	B	10	394.5 ± 32.1	48.5	0.0	0.0
Phi	C	4	253.5 ± 68.0	89.5	0.5	1.0
Phi	D	5	223.8 ± 105.0	105.4	0.0	0.0

Phi's no-discussion results show why avoiding breakdowns is only part of the task. Maintenance keeps the machine available, but each maintainer gives up a production opportunity. Once enough agents maintain to control wear, additional upkeep can reduce output without improving machine availability. Phi groups avoid breakdowns in both A and B, yet remain below the failure-free production reference.

Qwen's difficulty also involves timing. Its groups perform more maintenance than the 24-action threshold policies, but still experience breakdowns in every condition. A high maintenance total cannot compensate for a round in which everyone produces near the failure threshold. The group needs to allocate upkeep according to the current wear and the actions of its peers.

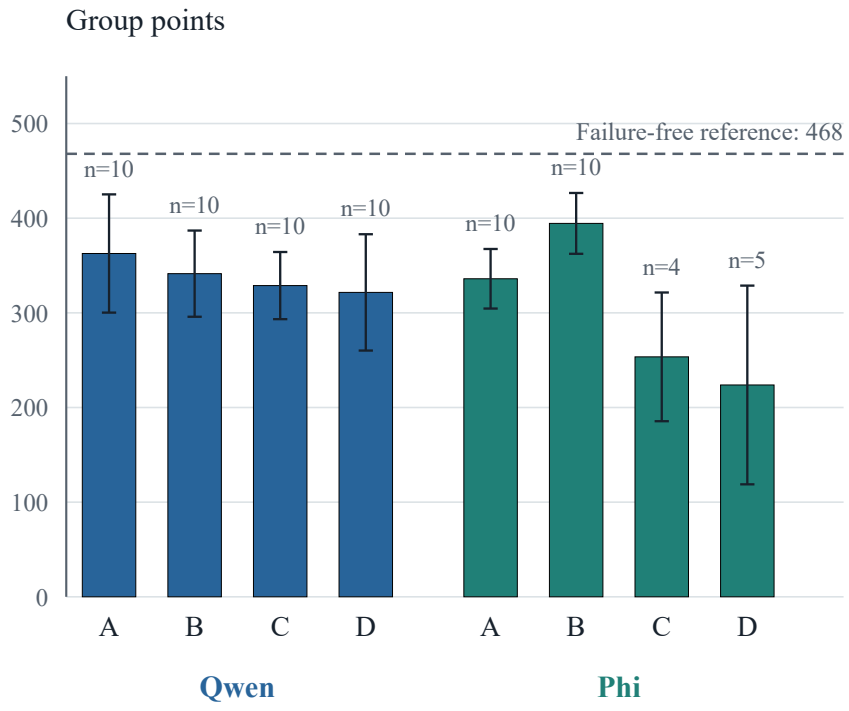


Figure 2: Group production by condition. Bars show episode means; error bars show one sample standard deviation. Labels give the number of finished workshops. The dashed line marks the 468-point failure-free reference.

5.2 Dashboard effects

The dashboard comparison asks whether summarising past contributions helps agents allocate the next maintenance action. All agents can inspect the action history. B and D provide the additional convenience of a per-agent

maintenance count. Table 5 shows how groups divide the work and how much maintenance capacity they leave unused.

Table 5: Maintenance distribution and waste in finished workshops. Concentration, score gap, and WWR are episode means. Unused capacity is total WWR divided by total supplied reduction, expressed as a percentage. Concentration excludes the one zero-maintenance episode in each of Qwen C and D.

Model	Condition	Concentration	Score gap	WWR	Unused capacity (%)
Qwen	A	0.232	16.2	177.6	66.2%
Qwen	B	0.278	18.9	174.6	73.1%
Qwen	C	0.289	26.4	183.4	74.6%
Qwen	D	0.274	21.9	213.3	76.6%
Phi	A	0.249	28.2	284.6	69.8%
Phi	B	0.288	30.0	152.1	52.3%
Phi	C	0.191	15.0	454.0	84.5%
Phi	D	0.188	11.4	550.6	87.1%

Phi benefits from this summary without discussion. Its groups perform less maintenance, waste less reduction capacity, and continue operating without breakdowns. Mean output rises from 336.0 to 394.5 points, and every matched B workshop earns more than its A counterpart. Because neither condition loses a round to shutdown, the gain comes from replacing maintenance with production. The summary may have helped agents avoid repeating work that others had already performed.

Higher output does not mean a more equal workload. In B, the most active maintainer takes a larger share of the group's upkeep, while the score gap widens. Phi groups reduce their total maintenance burden without spreading the remaining work more evenly.

Qwen does not show the same benefit. Its groups maintain less under B but experience more breakdowns, so the saved maintenance actions do not translate into higher output. Adding maintenance totals in the discussion conditions also provides no consistent production gain across the tested configurations. The summary helps in one setting; it does not supply a maintenance schedule or assign responsibility for the next round.

5.3 Discussion effects

Discussion gives agents a way to divide the work before acting. A useful exchange could identify a maintainer and leave the others free to produce. However, the agents submit nonbinding proposals and then choose their final actions in private. They must still turn the proposals into a compatible set of decisions.

The finished Phi discussion groups do not achieve this. They devote more actions to maintenance and produce less than the no-discussion groups. Their mean output even falls below the always-produce reference, which performs no maintenance and accepts repeated shutdowns. In these workshops, giving up production to maintain the machine costs more than the shutdowns incurred by that reference policy.

Qwen also shows no production benefit from discussion. Its groups continue to experience breakdowns despite exchanging proposals. Although their mean output is lower with discussion, variation between workshops leaves the direction of this comparison uncertain. Phi's comparison has a different limitation: only four C workshops and five D workshops finish. The following action patterns describe those finished groups.

5.4 Workload and waste

Phi's discussion groups come closer to an equal division of maintenance than its no-discussion groups. Their smaller score gaps also show that agents give up similar amounts of production. Yet they leave more than 84% of their supplied maintenance capacity unused. The agents share the burden of upkeep while performing far more work than the machine can use.

Fig. 3 places each workshop beside the reference schedules. A group can move toward higher maintenance counts without gaining more useful wear reduction, because the machine cannot become less worn than zero. Agents must therefore coordinate both who maintains and how many maintainers the machine needs.

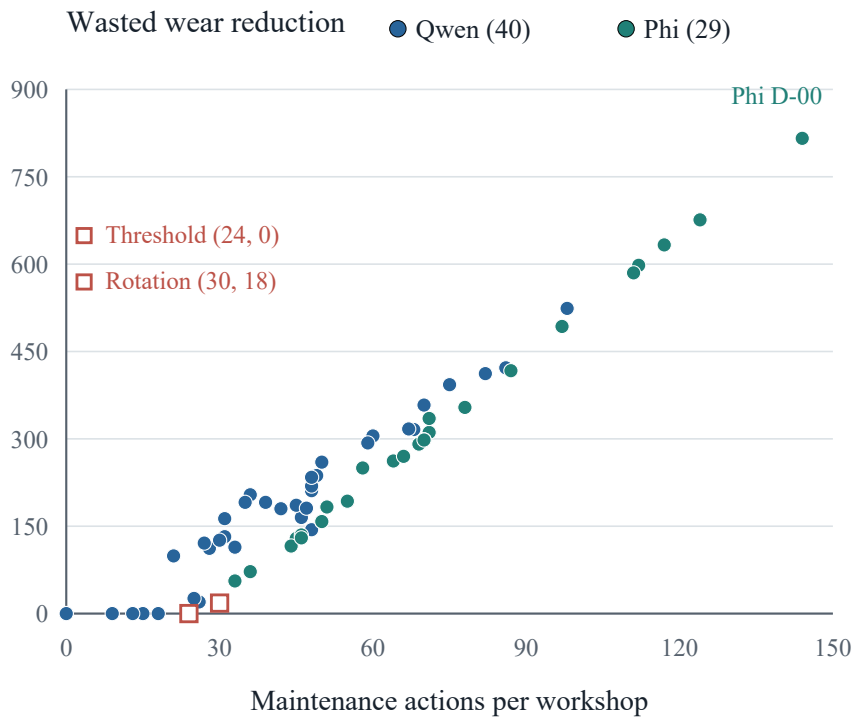


Figure 3: Maintenance actions and wasted reduction for each finished workshop. Blue points represent Qwen and green points represent Phi. Red squares mark the threshold and rotation reference policies. The axes do not show breakdowns or production, so a low-waste point need not represent a well-performing workshop.

The Phi D workshop `m1-D-00` illustrates the cost of failing to make that distinction. Round 3 starts with an unworn machine, but all six agents maintain. Nobody produces, and none of the supplied maintenance capacity can reduce wear. The agents repeat the same actions in the next round. They finish the workshop without a breakdown, but earn only 108 points after performing 144 maintenance actions.

Qwen shows a related scheduling problem in `m0-D-06`: the group performs extensive maintenance and wastes much of its capacity, yet the machine still breaks down. These examples explain why workload equality and maintenance quantity cannot establish whether a group has organised useful upkeep. We must also examine the machine's condition when agents act.

5.5 Intentions and actions

The proposal records help locate the coordination problem. Agents carry out most of their stated maintenance intentions, which might suggest that discussion produces stable plans. However, this measure overlooks agents who propose production and then switch to maintenance. Table 6 records both transitions.

Table 6: Proposal-to-action counts summed across finished discussion workshops. Fulfilment is the percentage of declared maintenance intentions followed by maintenance. The last column counts final maintenance actions that began with a production intention.

Model	Condition	Declared	Fulfilled	Rate	$P \rightarrow M$
Qwen	C	345	332	96.2%	78
Qwen	D	401	388	96.8%	76
Phi	C	80	68	85.0%	290
Phi	D	93	87	93.5%	440

Phi makes most of its final maintenance choices after proposing production. These switches account for more than four fifths of its upkeep in both discussion conditions. Qwen shows a smaller share of this transition, with most final maintenance actions following an intention to maintain.

One Phi D workshop makes the distinction clear. In `m1-D-08`, no agent proposes maintenance at any point, yet the group performs 51 maintenance actions. The proposals therefore give a different picture of the intended division of work from the final actions.

Under the protocol, agents can revise their intentions without confirming a shared assignment. Several participants can switch to upkeep in the same round, duplicating effort instead of leaving production to their peers. Phi's switches occur at this final-decision stage. They show where much of its maintenance enters the action sequence, although the records do not establish each agent's reason for switching.

5.6 Peer recognition

Recognition asks whether agents acknowledge peers who give up production to maintain the machine. A positive association would mean that heavier maintainers tend to receive more credit. Table 7 shows no common pattern across the two configurations and four conditions.

Table 7: Recognition analysis. Complete sets contain valid allocations from all six evaluators. Defined episodes have variation in both maintenance counts and received recognition. Mean Spearman reports the average correlation across those episodes.

Model	Condition	Complete sets	Defined episodes	Mean ρ
Qwen	A	9	9	-0.337
Qwen	B	10	10	0.018
Qwen	C	10	9	-0.165
Qwen	D	9	8	-0.188
Phi	A	3	3	0.256
Phi	B	5	5	0.283
Phi	C	2	2	-0.331
Phi	D	2	2	-0.271

Qwen's mean association is negative in most conditions and close to zero in B. Phi's association is positive without discussion and negative with discussion. These directions suggest that agents do not recognise maintenance effort in a consistent way. Phi's discussion estimates, however, come from only two complete recognition sets per condition.

Individual allocations also show that greater maintenance effort need not bring greater recognition. In `m1-A-01`, the heaviest maintainer receives less recognition than a peer who maintains less and earns more production points. The group does not compensate that participant's larger production sacrifice with greater acknowledgement.

Interpreting this as neglect of useful work would require more evidence. Maintenance counts include actions that remove no wear, so a larger count does not establish a larger useful contribution. The present results show a clearer problem in organising maintenance than in recognising it: groups can perform substantial upkeep, share it evenly, and still allocate it poorly.

6 Conclusion and Future Directions

The Maintenance Society examines how groups of small language model agents divide their effort between private production and shared equipment maintenance. Across our benchmark, the effect of displaying maintenance contributions differed between the two model configurations. For Phi, displaying maintenance totals reduced unnecessary upkeep without discussion, whereas Qwen continued to suffer repeated breakdowns regardless of dashboard information. Introducing discussion created a different coordination failure. Instead of organizing a sensible division of labor, finished Phi groups performed excessive maintenance, including rounds in which several agents maintained together. These dynamics demonstrate that equal participation is not a reliable indicator of collective efficiency, as agents can divide a maintenance burden with high equity while producing little of value.

These patterns may reflect both model tendencies and weaknesses in coordination. The models may have associated maintenance with cooperation and selected it without checking whether more upkeep was needed. The protocol provided no shared assignment. Agents submitted separate proposals and then chose their final actions independently. Several agents could therefore choose maintenance in the same round, duplicating one another's effort. Phi's high maintenance waste and few breakdowns are consistent with overly cautious action selection, while Qwen's combination of waste and shutdowns points to inconsistent maintenance timing.

Further experiments could test whether these failures depend on model capacity, agent prompts, or interaction rules. Testing larger models or architectures trained on quantitative tasks could show whether over-maintenance disappears with scale. Tweaking prompt framing, displaying wasted effort, or adding brief scratchpad steps might give agents better footing to track wear. The next step is to identify which changes help agents allocate useful maintenance while preserving production.

References

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A Paired Production Comparisons

Pairs use the same episode index and require both workshops to finish. The difference is the first condition minus the second. Intervals are descriptive 95% bootstrap intervals from 20,000 resamples of the observed pairs. They do not recover outcomes from unfinished workshops.

Table S1: Paired production comparisons among finished workshops. Positive differences indicate higher output in the first condition. Phi discussion comparisons contain fewer available pairs.

Model	Comparison	n	Mean Δ	95% interval
Qwen	B-A	10	-21.3	[-54.0, 7.8]
Qwen	D-C	10	-7.2	[-54.6, 32.7]
Qwen	C-A	10	-33.9	[-78.3, 11.1]
Qwen	D-B	10	-19.8	[-60.0, 24.0]
Phi	B-A	10	+58.5	[42.3, 75.6]
Phi	D-C	3	-54.0	[-183.0, 57.0]
Phi	C-A	4	-87.8	[-135.0, -40.5]
Phi	D-B	5	-169.2	[-225.6, -105.6]